RICE UNIVERSITY

Capstone: a controlled systems experiment

In pairs: two controlled experiments in one track, predicted with a cost model and explained with measurements or traces.

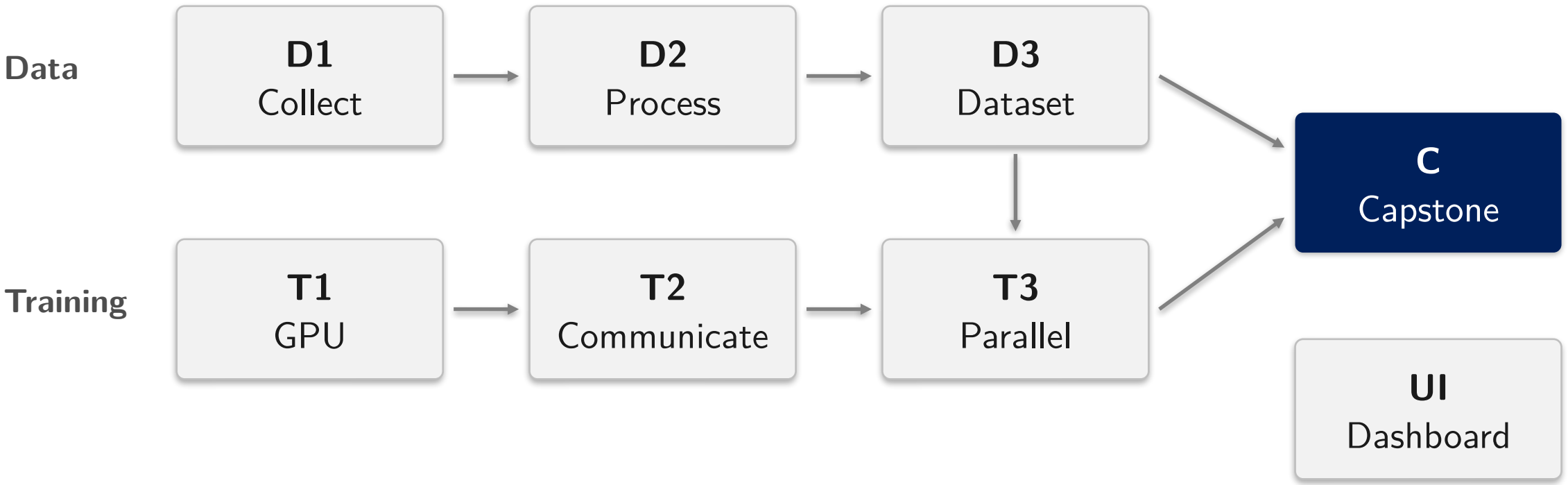
COMP 329 course staff

Rice University

Spring 2027 · Capstone

Spec week 7 · Pairs set week 13 · Due Tue May 4, 2027 · 13% of grade

Where this assignment fits



In our pipeline

This assignment: `D3 shards or D1 Parquet` → **C** → `comparisons.json + 6 pages`

What you will build, and why

Ask two clear systems questions in one track, change one thing at a time, predict with a cost model, and explain the result with evidence.

1. **No new architecture**

   The pipeline already runs; the work is the experiment design.

2. **Controls**

   The supplied runner refuses comparisons that change what the protocol fixes.

3. **Honesty**

   A well-supported negative result can earn full credit.

What goes in, what comes out

The shared pipeline
`D3 shards or D1 Parquet`
Staff components allowed

→

C
Capstone

→

Pair report
`comparisons.json + 6 pages`
Plus a 5-minute demo

Files you produce

`One-page plan`
Questions, predictions, metrics, controls

`comparisons.json`
Every repeat, medians, speedup, `controls_ok`

`Report + demo`
Cost model, measurement, trace, limitations

Two tracks: data systems or training systems

Data systems track

`partitioning`
`join or shuffle strategy`
`caching`
`data loading`

Fix dataset contents and model; verify equivalent results.

Training systems track

`kernel`
`compilation setting`
`communication schedule`
`parallel layout`

Fix model and data; control effective batch and precision.

Key idea

Predict each result with a cost model first. The provided protocol fixes validation data, baseline, budget and tolerances.

How a sound comparison is built

1 — Question
One change, named metric.

2 — Predict
Direction and cost, before running.

3 — Control
List what stays fixed.

4 — Repeat
One warmup + three measured runs.

5 — Explain
Uncertainty, confounders, limits.

Key idea

Keep validation data, baseline, budget and precision fixed; check model quality (training) or identical outputs (data).

Describe two questions; the runner checks them

```
TRACK = "training"
QUESTIONS = [
 {"name": "Q1: compile?",
  "baseline": {"mode": "eager"},
  "variant":  {"mode": "compiled"}},
 {"name": "Q2: fused kernel?",
  "baseline": {"mode": "eager"},
  "variant":  {"mode": "kernel"}}]
```

1. **Training settings**

   mode, `world_size`, implementation (ours or ddp).

2. **Data settings**

   stage (d2, d3), partitions, workers.

3. **Protocol checks**

   Same budget; validation within 0.01 bits/byte, or identical records.

4. **Repeats**

   One warmup, then three, alternating order.

What you write, what you get

You write

`Questions, hypotheses, predictions`

`starter/experiment.py` · TODO comments mark choices

`Plan and report`

`templates/` · `EXPERIMENT_PLAN.md`, REPORT.md

Provided

- `capstone_runner.py` protocol and checks
- T1 modes, T3 data parallelism, DDP
- Your D2/D3 Spark jobs (data track)
- T1 staff copy in provided/

Ground rules

Keep the signatures · don't edit `starter/pipeline_lab` · run through `course.py` or `lab.py`.

Build it in this order

1 2 3 4 5

Pick + plan

Track, two questions, one-page plan.

Smoke test

3 steps, 1 repeat.

Measure

One warmup + three repeats per side.

Analyze

Cost model, trace, limitations.

Demo

Raw input → shard → training → comparison.

```
experiment.py --smoke
```

```
experiment.py
```

Check

Done when: Two comparisons with predictions, `controls_ok`, three repeats each, and a causal explanation.

If something goes wrong

Two things changed at once
Check: One setting per question.

`controls_ok is false`
Check: The comparison cannot support a causal claim.

Only timings, no mechanism
Check: Use a trace, counter or cost model.

Warmup counted as a repeat
Check: The runner marks it; leave it out.

Two GPUs on one server called two servers
Check: Say what you actually ran.

Negative result hidden
Check: Report it and explain it.

The performance study

Does the change help, and why?

Predict first

Write the expected direction and size from a cost model: bytes moved, α–β time, bubble fraction, or shuffle size.

Training track
kernel, compile, schedule, layout

Data track
partitions, shuffle, caching, loading

How to measure: One warmup + three repeats per side, alternating order. Report performance and model quality or result equivalence.

How you are graded

25	25	35	15
Design	Reproducibility	Analysis	Communication

Capstone = 13% of the course grade · pairs · 6-page report + contribution statement

Check

Done when: Two comparisons with predictions, `controls_ok`, three repeats each, and a causal explanation.

Your first ten minutes

```
$ bash setup.sh
$ source .venv/bin/activate
$ python course.py preflight
$ python 07_capstone/starter/experiment.py \
    --input runs/inputs/bpe \
    --output runs/capstone-smoke --smoke
$ python 07_capstone/starter/experiment.py \
    --input runs/inputs/bpe \
    --output runs/capstone-r1
```

Your folder

`07_capstone/`

Handout:
`07_capstone/README.md`